

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 422 — Past Questions & Materials
by Renance

Course: CHM 422
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 400

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Paper 1 of 3 — CHM 422 - Radiochemistry and Nuclear Chemistry

Level 400 • Second Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

DEPARTMENT OF CHEMISTRY

SECOND SEMESTER EXAMINATIONS 2024/2025 ACADEMIC SESSION

CHM 422: RADIOCHEMISTRY AND NUCLEAR CHEMISTRY (2 Credits)

Instruction: Answer Question 1 and 2 others Time: 1% hours

Useful constants: ϕ

Avogadro's number = 6.023×10^{23} ; Mass of proton = 1.007277 amu

Speed of light (c) = 3×10^8 m/s; Mass of neutron = 1.008665 amu

6.022137×10^{23} amu = 1g; Mass of an electron = 0.0005468 amu

$1 \text{ kg} = 6.022 \times 10^{26}$ amu Mass of ^1_1H hydrogen atom = 1.007941 amu

$1 \text{ amu} = 931.5 \text{ MeV} = 1.660540 \times 10^{-27} \text{ kg}$; $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$

Question 1 (30 marks)

i) Iodine-131 has a half-life of 8 days. What percent of a sample would remain after 24 days? (3 marks)

ii) The isotope Bi-214 decays from 60.00 g to 7.50 g in 1 hour. Calculate its half-life. (3 marks)

iii) ^{23}Na decays to form ^{23}Ne . Write the equation for the reaction. (3 marks)

iv) If the half-life of ^{34}Na is 2.6 years, after how many years will a sample decay to 12.5% its original amount? (3 marks)

v) The measured mass of neutral Li-6 is 6.01512 amu. What is the mass defect of this isotope? (3 marks)

vi) What is the binding energy in ^4He nucleus if its mass defect is 0.08181 amu? (3 marks)

vii) List three applications of radioactive isotopes giving one example in each case. (3 marks)

viii) List the three common particles and their symbols used in describing of radioactive decay and nuclear transformations. (3 marks)

ix) Complete the following equations and then write them in short notation. (3 marks)

a) $^{13}\text{N} + ^1_0\text{n} \rightarrow ^{13}\text{C} + ^1_1\text{H}$

b) $^{238}\text{U} \rightarrow ^{234}\text{Th} + ^4_2\text{He}$

x) The activity of a radioactive isotope reduces by 25% after 100 minutes, Calculate the decay constant and half-life period. (3 marks)

Question 2 (20 marks)

a) A freshly cut piece of plant gives 20.4 counts per minute per gram. A piece of wood antique gives 12.18 counts per minute per gram, What is the age in years of the antique? It is assumed that the radioactivity is entirely due to C-14. The half-life period of C-14 is 5760 years. (10 marks)

b) When stable, non-radioactive nitrogen in the air is subjected to neutron bombardment during an H-bomb explosion, a nuclear reaction occurs, producing protons. Write a balanced nuclear equation

for this reaction and name the other product of the reaction. If the other named product is radioactive

and undergoes beta decay, write an equation for the decay process and identify the new daughter element. (5 marks) ;

ϕ) Calculate the neutron-to-proton ratio for each of the following atoms: (i) The mass number and atomic number of a radioactive element, Actinium, are 227 and 89 respectively (ii) carbon-12 (iti) mass number and atomi

ae mic number of

potassium-39. (5 marks) of the element, lead, are 207 and 82 respectively (i
ely (iv) carbon-14 (vj

Question 3 (20 marks)

a) Compare and contrast the following:

(i) Nuclear reactions and chemical reactions

(ii) Nuclear fission and nuclear fusion

(iii) Electron and positron. (5 marks)

BOR ane complete the table below (5 marks)

uclear Equation Name of decay process "Type of radio:

Sc ___Name of decay process___'Type of radioactivity

i. | $^{49}_{19}\text{K} + 2\text{TBAT}$

ii $\text{MW} + ? + \text{HER}$

i $^{238}_{92}\text{U}$

iv $^{738}_{44}\text{U} \rightarrow ^{240}_{94}\text{Pu} + ?$

—_ i a

v. $\text{FAL} + 2\text{-}^7_{11}\text{B} + \text{An}$

290 amu. Find each of the following.

c) The experimen tally determined mass of @Mn is 52.941

(i) The mass defect

(ijThe nuclear binding energy

inding energy per nucleon (10 marks)

(iii) The nuclear b

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bundance of

Question 4 (20 marks)

ds as KCl] and is an essential

40 has a natural 4

is by beta

a) Potassium ion is present in foo

occurring isotopes of potassium potassium-40, is radioactive.

0.0117%- i undergoes radioactive decay in three ways: . by electron capture, 1.35% emission, and 0.49% is by positron emission.

which ^{40}K decays.

ree modes by

decay? (5 marks)

ear equations for the th

u% of the ^{40}K in a sample to

CiyW rite the nucl

oes it take for 1.0

9 [sotopes-R-US for

until 5:00

receiving the compound?

nt. (7 marks)

(ii) How long d

b) A hospital nee

709 mg of $^{22}\text{RaCl}_2$ at 11:00 pm. 7

pm, Tuesday. How much should the Radiologist pa

Market value of ^{22}Ra ($T_{1/2} = 20.8 \text{ hr.}$) is \$

bjected to carbon-

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4.68 disintegrations/min per gram of

f 15.3 disintegrations/min per

original Rembrandt?

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c) An oil painting al

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Paper 2 of 3 — CHM 422 - Radiochemistry and Nuclear Chemistry

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1 kg = 6.022×10^{26} amu, Mass of ^1_1H hydrogen atom = 1.007941 amu

1 amu = 931.5 MeV = 1.660540×10^{-27} kg; 1 eV = 1.602×10^{-19} J

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iii) $^{24}_{11}\text{Na}$ decays to form $^{24}_{12}\text{Mg}$. Write the equation for the reaction, (3 marks)

iv) If the activity of a sample is 6.0 Ci, after how many years will a sample decay to 12.5% its original amount? (3 marks)

5 marks

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7 — ee ee oe tly 9; What is the mass de-

v) The measured mass of neutral Li-6 is 6.01512 amu. What is the mass defect of this isotope? (3 marks)

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: ' of How he 0 ahs Pale ; ,

vi) What is the binding energy in ^6_3Li nucleus if its mass defect is 0.08181 amu? (3 marks)

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ve Chi lege LW TE Slyeocidiz (AME aR Re ean. 4if : "

viii) List the most common particles and their symbols used to describe radioactive decay and

nuclear transformations, (3 marks)

ix) Complete the following equations and then write them in short notation. (3 marks)

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b) ${}^{137}_{54}\text{Xe} \rightarrow {}^{137}_{55}\text{Cs} + {}^0_{-1}\text{e}$

we Qrtan . if

} ; j ;

c) ${}^{165}_{64}\text{Gd} \rightarrow {}^{165}_{65}\text{Eu} + {}^0_{-1}\text{e}$

x) The activity of a radioactive isotope reduces by 25% after 100 minutes. Calculate the decay constant and

half-life period. (3 marks) ha Ho UH -f Ha 5%, —= & CHr 99D, 4 Hoo

4 ,

${}^{20}\text{H} + {}^{16}\text{O} \rightarrow {}^{36}\text{S} + {}^4\text{He}$

Question 2 (20 marks) Pet, OS0- H Beds soi,

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and undergoes beta decay, write an equation for the β^- process, and identify the new daughter element. (5 marks) Hyd ogera tins OF De ding W3diagtery Mycol tr ter teinin

c) Calculate the neutron-to-proton ratio for each of the following atoms: (i) The mass number and atomic number of a radioactive element, Actinium, are 227 and 89 respectively (ii) carbon-12 (iii)

${}^{12}_6\text{C}$ — 07 €- Conn bg ct, CH, —0-E- Cote CH,

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(Hy 7 O- é. COs 4 CH, } ff

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mass number and atomic number of the element, lead, are 207 and 82 respectively (iv) carbon-14 (v)

potassium-39. (5 marks), a) ADA E22 C18 | tq}

Orewa C

Question 3 (20 marks) ANOS Ri ernot

f — Va aste} : <1

a) Compare and contrast the following: aise 1) a ORY

(i) Nuclear reactions and chemical reactions. / | pool!

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JG \ — Iyaolere acu (19:2(9, 0

(ii) Nuclear fission and nuclear fusion.

(iii) Electron and positron. (5 marks) PIAIAW Ma cackidonte = C20. (54,0

it eae |

b) Copy and complete the table below (5 marks)

S/N Nuclear Equation ___ Name of decay process _ Type of radioactivity 7

i, e+ 2 tar CHa -O-£— R Cth - OH

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iii, 189Wt 2 — TERE cH -o-0 -

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IV “a0 — eT hh + ¥ CHe ao -¢ a ? ¢ Ht - OH

. 3 { A Q wy et ar ea i

c) The experimentally determined mass of ^{3}Mn is 52.941290 amu. Find each of the following.

(i) The mass defect Pal. Weeee fable. — Put Meqeadafirs

(ii)The nuclear bindi . = Gr) ve (ot) °

ii) The nuclear binding energy « mab \ (qi? 25

= as (chy OUD, ai CCHy 0H y dct

(iii) The nuclear binding energy per nucleon (10 marks) } !

(Hod b CHg0H cH

Question 4 (20 marks)

a) Potassium ion is present in

occurring isotopes of potassium, po

0.0117%, It undergoes radioactive decay in tl

emission, and 0.49% is by posiirun Cinisstan. |

foods as KCl and is an essential nutrient in the human body. One of the naturally 4,

tassium-40, is radioactive. Potassium-40 has a natural abundance of te

aree ways: 98.2% is by electron capture, 1.35% is by beta

Leek

(i)Write the nuclear equations for the three modes by which “°K decays. Cour

e to undergo radioactive decay? (5 marks) CH Ol

(ii) How long does it take for 1.00% of the *°K ina sampl

‘The radiologist places an order to Isotopes-R-Us for

hospital does not receive the compound until 5:00

inpany upon receiving the compound?

d on radium content. (7 marks)

b) A hospita! needs radiun: for chemotherapy.

709 ing of -3RaCh at 11:00 pm, Monday, but the

pm, Tuesday. How much should the Radiologist pay the co
Market value of “Ra (ti = 280.8 hr.) is \$50,000 per gram base
dealer to be painted by Rembrandt (1606-1669) is subjected to carbon-
painting shows an average of 14.68 disintegrations/min per gram of
ms today undergoes an average of 15.3 disintegrations/min per
Id this painting be an original Rembrandt?

c) An oil painting alleged by a
14 dating. The canvas of the
carbon. The carbon in living organism!
gram. If the half-life of carbon-14 is 5730 years, calculate
Show your calculations to prove your answer. (8 marks)
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Paper 3 of 3 — CHM 422 PQ

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